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Building facade system and method of forming a building facade

Abstract

A building facade system which comprises an anchor. The anchor having a flange extending in a first horizontal direction and serrations along a portion thereof. A vertical mullion being held in shear and secured by shear connection to the anchor to support horizontal members forming a frame for supporting a building facade panel. The shear connection formed by a fastener coupling the anchor to an outside lateral surface of the vertical mullion. The fastener extending in the second horizontal direction through the anchor and the vertical mullion. The angle member having first and second flanges having proximal ends joined together and opposing terminal ends. The second flange being engageable with the flange of the anchor by way of a vertically-oriented coupling securing the flange of the anchor and the angle member. The angle member being securable to a building floor slab and supporting the anchor thereon.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional application of U.S. patent application Ser. No. 17/847,986 filed on Jun. 23, 2022, now allowed, which is a continuation application of U.S. patent application Ser. No. 17/150,713 filed on Jan. 15, 2021, which issued as U.S. Pat. No. 11,396,750 on Jul. 26, 2022, which claims priority to U.S. Provisional Patent Application No. 63/064,194 filed Aug. 11, 2020 and U.S. Provisional Patent Application No. 63/055,300 filed Jul. 22, 2020. This application claims priority from each of the above-referenced applications, the disclosures of which are each hereby expressly incorporated by reference in their entirety as if set forth fully herein.

FIELD

(1) Embodiments presented herein relate generally to the field of building facade systems which form an envelope of external facade around buildings such as multi-residence or commercial office buildings, high-rise buildings, towers, skyscrapers and the like. More particularly, embodiments disclosed herein provide a universal building facade system anchored from the building floor

structure via a shear supported unified anchor innovation. According to exemplary embodiments, the building facade system presented herein requires fewer field installed parts than conventional facade systems and increases labor efficiency of installation while concurrently providing the ability to apply a traditional fire stop and smoke seal with a notched vertical configuration as required for the safety of building occupants and to meet international and local building codes after installation of the frame onto the floor structure.

BACKGROUND

(2) Two conventional types of building facade systems that are generally known and commonly used are window/hybrid wall and curtainwall. Generally, known curtainwall framework employs a plurality of anchor sub-assemblies. Each subassembly is comprised of roughly half of a two-part large aluminum mating clip, and can include a Jack bolt, and serrated washer. In assembling such systems, one subassembly is typically pre-attached to the building terminal slab end with a first crew of laborers and the second subassembly is mated to the pre-glazed panel by a second crew of laborers. The two subassemblies that make up the whole anchor are joined together when a third crew of laborers joins the pre-glazed panel anchor subassembly installed by the second crew to the subassembly that was attached to the floor slab by the first crew. The pre-glazed panel of such systems can have a plurality of anchor parts attached structurally to vertical structures/mullions in a shear or tensile vector. Both known curtainwall notched and unnotched vertical framework types stop short of interfacing the system with the building floor structure by over an inch, or as much as several inches. Such arrangement unfortunately has been shown to provide a direct fire path between floors within twenty (20) minutes after the fire burns through aluminum horizontals. As such, known curtainwall configurations can present a life safety hazard by allowing vertical fire spread if costly fire stop materials/measures are not added. Other notched curtain walls rely upon a continuous shelf held in tensile which prevents the field application of this traditional critical life safety fire stop measure in the field. Apart from critical fire safety limitations, the unprotected gap allowed by notched curtainwall systems allows for excessive sound to travel upwards to the occupants above.

(3) Some existing curtainwall systems utilize a continuous shelf design. In such designs, a traditional two-hour rated fire stop and a smoke seal may not be able to be installed in the field for the safety of the building occupants which can represent a safety hazard if used on a building. Such a limitation is critical with regard to the issue of firestopping between floors. For the safety and health of the building occupants building codes generally require the implementation of separate firestopping measures, such as fire resistive mineral wool and smoke resistant silicone seals be installed after the panel is affixed to the building to prevent fire and smoke from traveling up the curtain wall between floors. The installation and use of such measures can be expensive, time consuming, and may not be possible with certain curtainwall systems that utilize a continuous-shelf design due to the single shelf anchor spanning between vertical members.

(4) By contrast, window wall systems are generally known to be endo bearing fenestration systems provided in combination assemblies and composite units, including transparent vision panels and/or opaque glass or metal panels, which span from the top of a floor slab to the underside of the next higher floor slab—using the below floor slabs as structural support. Window wall system are load bearing directly on the floor slab and is comprised of any number of individual completed window units used to fill a particular opening on a particular floor. Thus, when a window wall system is fully installed within an opening in a building the system performs independently of other window wall systems in the building.

(5) Conventional building window/hybrid wall framework is generally known to employ a plurality of parts comprising a series of site-installed track parts at the top and bottom face of the terminal end of a floor slab to create a confined opening by two crews of laborers. A third laboring crew then insulates and covers the slab edge using a plurality of site installed loose shipped parts. A fourth crew installs pre-glazed units within the confines of the top and bottom track system set by

the first laboring crew. Thus, the installation process can be labor intensive. Further, since window walls are endo bearing, the glass aesthetic design is less continuous and more interrupted. Window walls can also be more susceptible to leaking due to the seals around the panels drying out.

(6) In view of the troublesome deficiencies of known curtainwall systems, there is a need in the art for a building facade system that is able to provide improved safety code compliant firestopping and smoke sealing capabilities, as well as better noise reduction, without requiring excessive installation and/or maintenance time and expense. Innovations presented herein, including the use of the unified vertical shear blade anchor, overcomes such deficiencies and eliminates the need for crews associated with preplacement of anchors required in common curtain wall systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a partial front elevation view of a building facade system according to exemplary embodiments provided herein.
- (2) FIG. 2 is a schematic cross-section elevation view of a portion of a building facade system according to exemplary embodiments provided herein taken along line A-A of FIG. 1.
- (3) FIG. 2A is a schematic detail cross-section elevation view of a portion of the building facade system shown in FIG. 1.
- (4) FIG. 3 is a schematic top plan cross-sectional view of a portion of a building facade system according to exemplary embodiments provided herein taken along line B-B of FIG. 1.
- (5) FIG. 3A is a schematic detail top plan cross-section view of a portion of the building facade system shown in FIG. 2.
- (6) FIG. 4 is a schematic cross-section elevation view of a portion of a building facade system according to exemplary embodiments provided herein taken along line C-C of FIG. 1.
- (7) FIG. 5 is a schematic perspective view of an exemplary angle member assembly according to embodiments provided herein.
- (8) FIG. 6 is a schematic perspective view of an exemplary vertical shear blade anchor according to embodiments provided herein.
- (9) FIG. 7 is a schematic perspective view of an exemplary anchor assembly according to embodiments provided herein.
- (10) FIG. 8 is a flow diagram of illustrating exemplary steps of a method for installing a building facade system according to embodiments provided herein.

DETAILED DESCRIPTION

- (11) While the subject invention is susceptible of embodiment in many different forms, there are shown in the drawings, and will be described herein in specific detail, embodiments thereof with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the invention to the specific embodiments illustrated.
- (12) Embodiments disclosed herein are generally directed to a building facade system and method of forming a building facade system substantially as shown and/or described in connection with the figures and as set forth more fully in the claims. It will be understood from the subject disclosure that embodiments presented herein can allow for the floor slab of a building structure to interface more closely with the interior of the building facade system by way of a unified vertical shear blade and an open ended or closed notch within a vertical mullion. It will be appreciated that the disclosed embodiments present an entirely new type of building facade system which provides for the application of fire stop measures as required for the safety of building occupants and also to meet international and local building codes after installation of the frame onto the floor structure. It will further be appreciated that disclosed embodiments provide a highly variable building facade that is

practically universal in application. Specific advantages, aspects and novel features of the disclosed system and method, as well as details of the illustrated embodiments thereof, will be more fully understood from the following description and drawings which reference specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the present invention.

(13) With reference now to the figures, FIG. 1 schematically illustrates a portion of a building facade system **10** constructed in accordance with embodiments provided herein. As shown schematically in FIG. 1, according to exemplary embodiments, building facade system **10** can be comprised of a plurality of unified panel assemblies **11** comprising at least one building facade panel **22**, **24** that can be structurally glazed onto a frame assembly comprised of vertical mullions **14** and horizontal members **16**, **18**, **20**. As shown schematically in FIG. 1, unified panel assemblies **11** can be arranged side-by-side along a portion of the exterior of a building structure to form a building facade. According to exemplary embodiments, the building facade can be comprised of substantially vertical mullions **14** and horizontal members **16**, **18**, **20** supported on vertical mullions **14**. Building panels **22**, **24** can be aligned both vertically and horizontally side-by-side and end-to-end with seals **72** therebetween to protect the and insulate the interior of the building from precipitation, wind and temperature.

(14) FIGS. 2-4 schematically illustrate portions of a building facade system **10** according to exemplary embodiments presented herein and schematically illustrated in FIG. 1. According to exemplary embodiments shown schematically in FIGS. 2-4, the building facade system **10** is shown as being comprised of unified panel assembly **11** installed to the terminal end of a building floor slab FS and can generally comprise an anchor assembly **12**, a vertical mullion **14**, horizontal members **16**, **18**, **20** and building facade panels **22**, **24**. As shown schematically in FIGS. 2-4, building facade system **10** can further comprise an interior trim assembly **90** shown as a floor closure sub-assembly as well as firestopping measures **26**, smoke seals **28** and associated fasteners, gaskets, seals, insulation, spacers as will be described further herein.

(15) As best seen in FIGS. 3 and 3A, an exemplary anchor assembly **12** can be comprised of a plurality of unified vertical shear blade anchors **32** laterally spaced-apart from one another and secured to opposing sides of a vertical mullion **14**. In particular, vertical mullion **14** can be comprised male and female mullion half members **14a**, **14b** that are generally rectangular shaped in cross section and securely snapped together to form vertical mullion **14**. According to exemplary embodiments, a unified vertical shear blade anchor **32** can be secured to each mullion half member **14a**, **14b**. As is conventionally known, the vertical mullion **14** and horizontal members **16**, **18**, **20** together form a frame **21** for supporting the building facade panels **22**, **24**. The frame, and frame components and hardware can largely be comprised of extruded aluminum, although other materials can also be used without limitation. A back pan **59**, such as a galvanized steel back pan, can be sealed to the frame **21** on all sides. As best shown schematically in FIGS. 2 and 4, a portion of the vertical mullion halves **14a**, **14b** can be provided with a notched section **15** to minimize the distance of the wall of the facade from the terminal face of the floor slab FS and to allow incidental building movements and thermal expansion without compromising the integrity of the building facade. The notched sections **15** can extend along a portion of the length of the vertical mullion and permit the frame **21** to be positioned closer to the terminal edge of the floor slab and provide a space for the application of a fire stop **26** and smoke seals **28**.

(16) From the subject disclosure it will be readily understood by persons of ordinary skill in the art that FIGS. 2-4 are schematic illustrations of an exemplary anchor location that can be part of a much larger building facade system **10**. In particular, it is generally known that the overall system can encircle an entire exterior of a building structure, or large portions thereof, to span multiple floors to form an exterior facade for the building. Thus, persons of ordinary skill in the art will

recognize and appreciate that the portion of the building facade system **10**, anchor assembly **12**, and other components shown in the FIGS. **2-4** can be provided in pluralities and at numerous locations around the exterior of a building structure in an ordered arrangement. From the subject disclosure it will further be recognized that the building facade system **10** shown and described herein can be comprised of unified panel assemblies **11** that can be shop assembled and require no pre-attachment of anchors to a building floor structure. Such unified assemblies **11** according to exemplary embodiments can generally comprise unified vertical shear blade anchor **32**, frame assembly **21** comprised of mullions **14** and horizontal members **16, 18, 20** and a building panel **22, 24** structurally glazed onto the frame assembly.

(17) As illustrated schematically in FIGS. **2-4**, according to exemplary embodiments the anchor assembly **12** can be generally comprised of a unified vertical shear blade anchor **32** and angle member **40**. FIG. **5** illustrates the unified vertical shear blade anchor **32** according to exemplary embodiments presented herein. As shown schematically in FIG. **5**, unified vertical shear blade anchor **32** can have a body portion **33** and a flange **34** extending horizontally from the body portion in a first direction. According to exemplary embodiments, flange **34** can have a proximal end adjacent body portion **33** and an opposing terminal end and opposing top and bottom surfaces. The bottom surface can have downwardly projecting serrations **35** along at least a portion thereof. Serrations **35** can have a sawtooth-type arrangement comprised of a pattern or series of alternating elongated ridges and grooves; the ridges and grooves extending in a second direction across at least a portion of the width of flange **34**. As shown in FIG. **5**, flange **34** can have a tab **36** extending laterally from a side edge of the main flange section. Tab **36** can be rectangular shaped and extend along at least a portion of the side edge of the main flange section extending all the way to the terminal end of flange **34** as shown in FIG. **5**. Flange **34** can also have an opening **37** extending therethrough between the top and bottom surfaces. As shown in FIG. **5**, opening **37** can have an elongated or slotted shape having a length extending in the first direction.

(18) According to exemplary embodiments shown schematically in FIG. **5**, the body portion **33** of unified vertical shear blade anchor **32** can extend downward from the proximal end of flange **34** and have a top portion adjacent the flange **34** and an opposing bottom portion. As shown in FIG. **5**, body portion **33** can slope away from flange **34** as it extends from top to bottom such that the body portion and flange extend away from one another to define an obtuse angle below the flange. As shown in FIG. **5**, holes **38** can extend through body portion **33** in a second direction substantially perpendicular to the first direction. As described in further detail below, holes **38** can be configured for receiving fasteners for securing the unified vertical shear blade anchor **32** in shear to a vertical mullion half section **14a, 14b**. As shown in FIG. **5**, the holes **38** can have a diameter and circumference only slight smaller than the width of the body portion **33** and the top and bottom portions of the body portion can have bulbous protrusions **39a, 39b** for providing sufficient surrounding area to accommodate holes **38**. According to exemplary embodiments provided herein, unified vertical shear blade anchor **32** can be made of extruded aluminum, although it will be understood that it can also be made from other rigid materials without limitation, such as galvanized steel for example.

(19) From the subject disclosure, it will be generally understood and appreciated by persons of ordinary skill in the art that the invention and utilization of a unified vertical shear blade anchor **32** in accordance with embodiments presented herein creates an entirely new variant of building facade systems that is universal in application. Specifically, such innovation can provide the aesthetic contemplated by all prior types of building facade enclosure systems described above in a single system and further provides dramatically improved design freedom within a single unified chassis. This is made possible by the encapsulation of the floor slab that is made possible by the combination of functions between the notch and the innovation of unified vertical shear blade anchor **32**. Such capabilities and improvements can be obtained without the need for multiple laboring crews to mate curtainwall framework anchors of the type used with prior curtainwall

systems because the frame contains within itself all the required anchor components and eliminates the need to pre-attach anchors to the building while also allowing the installer to install the needed fire safety systems after the frame is affixed to the building. Such capability is not achievable with any known notched curtainwall which instead rely on a single shelf anchor holding the unit in tensile.

(20) FIG. 6 illustrates an exemplary angle member **40** according to embodiments presented herein. As shown schematically in FIG. 6, angle member **40** can generally have an 'L'-shaped configuration formed by substantially perpendicular first and second flanges **42**, **44** which set apart from one another at an angle on the order of 90 degrees. According to exemplary embodiments shown schematically in FIG. 6, the first and second flanges **42**, **44** can be joined together at their respective proximal ends and can each have an opposing terminal end. As illustrated in FIG. 6, the first and second flanges **42**, **44** can be sized differently with the first flange **42** having a longer length than second flange **44**. It will be understood however that the flanges **42**, **44** can have the same dimensions or can have additional differences, such as different thicknesses or weights and that the sizes, dimensions or other properties of the flanges can be varied to accommodate different loads and floor slab construction tolerances as need be. According to exemplary embodiments provided herein, angle member **40** can be made of extruded aluminum, although it will be understood that it can also be made from other rigid materials without limitation, such as galvanized steel for example.

(21) As shown in FIG. 6, second flange **44** can have a top surface with upwardly projecting serrations **46** along at least a portion thereof. Serrations **46** can have a sawtooth-type arrangement comprised of a pattern or series of alternating elongated ridges and grooves; the ridges and grooves extending in the second direction across at least a portion of the top surface of angle member **40**. As described in further detail below, the upwardly projecting serrations **46** of angled member **40** are configured for engagement with the downwardly projecting serrations **35** of the unified vertical shear blade anchor **32**. The second flange **44** can also have a threaded opening **48** extending therethrough between the top and bottom surfaces. As shown in FIG. 6, opening **48** can have a generally circular or cylindrical shape, but it will be understood that it can have additional shapes without departing from the scope of embodiments presented herein.

(22) Returning to FIGS. 2-4, according to exemplary embodiments, an elongated fixture such as a steel channel or tube C can be permanently preplaced onto or embedded into the building floor slab FS by means of welding or casting in place. As shown schematically in FIG. 2, channel C can be secured adjacent the terminal end of the floor slab FS such that the outside edge of channel C is flush with the terminal edge of the floor slab and the top surface of channel C is flush with the top edge of the floor slab FS. According to exemplary embodiments shown schematically in FIGS. 2-4, the terminal end of the first flange **42** of angle member **40** can be inserted into an interior portion of elongated channel C and secured therein. Indexing angles **50** can be installed alongside angle member **40** to laterally index angle member **40** within channel C. In such orientation, the first flange **42** of angle member **40** will extend in a substantially vertical direction and the second flange **44** will extend in a substantially horizontal direction as shown in FIGS. 2-4. Indexing angle **50** can be comprised of extruded aluminum or another rigid material.

(23) Engagement of angle member **40** within channel C can serve as a windload anchor in lieu of providing and/or relying on a leveling bolt to extend to the bottom of the channel to act as both a windload and deadload design. The use of angle member **40** in this manner represents a dramatically improved anchor design for building facade systems. For example, such arrangement provides greater surface area contact to improve rotational force and improved performance under seismic loading with easier pinning as needed. Such design can additionally reduce vertical eccentricities from centroid that make the anchor more structurally efficient along the vertical "up-down" adjustable axis. The "L"-shaped wind loaded anchor angle can further act as a compressioned composite when tightened by the female-type fastener to also reduce the horizontal

eccentricities from centroid which can make the anchor more structurally efficient along the lateral “in-out” adjustable axis.

(24) According to exemplary embodiments shown schematically in FIGS. 2-4, the body portion **33** of unified vertical shear blade anchor **32** can be secured by a shear connection to a vertical mullion **14**, and more particularly to the outside lateral surface of a vertical mullion half **14a**, **14b**, by fasteners such as shear bolts **52**. According to embodiments presented herein, shear bolts **52** can extend in a second direction and be inserted into holes **38** in the body portion of the unified vertical shear blade anchor **32** and fastened to vertical mullion **14** with associated fasteners. Such attachment can include a bearing insert **54** to attach the shear blade anchor to the vertical mullion **14**. Bearing insert **54** can be comprised from extruded aluminum or other ridged material without limitation. Thus, according to exemplary embodiments presented herein, vertical mullion **14** can be secured by shear connection to the unified shear blade anchor **32** with vertical mullion **14** being held in shear to suspend the frame **21** assembly from the building floor slab FS.

(25) As shown in FIGS. 2 and 4, upon installation of a unified anchor assembly and alignment adjacent to a floor slab FS, flange **34** can extend inward and above the terminal edge of floor slab FS with serrations **35** along the bottom side of flange **34** extending downward. According to exemplary embodiments shown schematically in FIGS. 2, 4 and 7, downwardly projecting serrations **35** on vertical shear blade anchor **32** can engage upwardly projecting serrations **46** on the top surface of angle member **40**. Such engagement can form anchor assembly **12** and can secure the unified vertical shear blade anchor **32** in the horizontal direction and perpendicular to the terminal edge of the floor slab FS. Anchor assembly **12** is further illustrated in FIG. 7, and engagement between the serrations of angle member **40** and unified vertical shear blade anchor **32** can further support the frame **21** in the desired horizontal position relative to the said floor slab FS and channel C welded onto or cast into the floor structure. More particularly, the relationship of these serrated members, together with the slotted opening through the flange **34** of unified vertical shear blade anchor **32** can permit horizontal in-and-out adjustment of the frame relative to said channel C.

(26) According to exemplary embodiments shown schematically in FIGS. 3 and 3A, a plurality of unified vertical shear blade anchors **32** are shown as being secured adjacent one another to mullion **14**. As illustrated in FIGS. 3 and 3A, the plurality of unified vertical shear blade anchors **32** can be secured adjacent one another to opposing mullion half members **14a**, **14b** with the tabs **36** of the flanges **34** of unified vertical shear blade anchors **32** being positioned along an interior side or face of mullion half members **14a**, **14b**. Thus, the tab **36** along the interior surface of flange **34** of each shear blade anchor can form a notched portion to accommodate the vertical mullion half **14a**, **14b**. As shown schematically in FIGS. 3 and 3A, according to exemplary embodiments a gap or space can be provided between adjacent unified vertical shear blade anchors **32**, and more particularly between interior surfaces of flanges **34** between tabs **36**. The gap can be provided for accommodating a sealant **56** as shown in FIG. 3A.

(27) According to exemplary embodiments shown schematically in FIGS. 2, 3 and 7, openings **48** in angle members **40** can be aligned with the slotted openings of unified vertical shear blade anchors **32** and fasteners such as, for example, threaded leveling bolt anchors **60** can be provided and inserted through the slotted openings in the unified vertical shear blade anchor **32** and threaded through the threaded opening **48** of angle member **40**. As shown schematically in the figures, leveling bolt anchors **60** can be provided with corresponding female-type fasteners such as, for example, high strength serrated flange locknuts **62** which can be threaded upon and secured to leveling bolt anchors **60** above the top surface of the flanges of unified vertical shear blade anchors **32**. Fasteners or leveling bolt anchors **60** can extend in a third direction through the flanges **40** of unified vertical shear blade anchors **32** and the second flanges **42** of angle members **40**. As shown schematically in FIGS. 2-4, the terminal ends of leveling bolts **60** can be seated upon a top surface or flange of channel C or a separate plate or angle bracket seated upon the top surface of the channel C. According to exemplary embodiments, the turning of leveling bolt anchors **60** against

the top of the channel C can allow for vertical adjustment of the unified vertical shear blade anchors **32** which can also commensurately move the frame **21** in positive or negative elevation from an initial nominal placement of the frame relative the floor structure FS.

(28) According to exemplary embodiments, the distance from top of said leveling bolts **60** to the top surface of channel C can be fixed. As shown schematically in FIGS. **2-4**, the unified vertical shear blade anchor **32** can rest on top of said angle member **40** which can be engaged to the top portion of channel C. The female-type fasteners can be adjustable up and down bearing on threads on said leveling bolts **60** thereby adjusting said unified vertical shear blade anchor **32** by moving the said below angle member **40**.

(29) According to exemplary embodiments shown schematically in FIG. **2**, a plurality of horizontal members **16**, **18**, **20** can be supported from vertical mullions **14**. Such horizontal members can be made from extruded aluminum or other rigid materials without limitation and can form frame **21** for supporting building facade panels **22**, **24** which can be structurally glazed to frame **21** and delivered to a building site as a prefabricated unified anchor panel assembly **11**. Horizontal member **16** can comprise, for example, an extruded aluminum head member secured to vertical mullion **14** in an area adjacent or around unified vertical shear blade anchor **32**. Horizontal member **20** can comprise, for example an extruded aluminum sill member secured to a lower portion of a vertical mullion **14** and above a head member **16** supported from a below floor slab FS. According to exemplary embodiments, horizontal sill member **20** can serve as the top portion of a windload connection load path for the frame below. Horizontal member **18** can comprise, for example, an extruded aluminum spandrel panel support member secured along the length of vertical mullion **14** below head member **16** and above horizontal sill member **20**. As shown schematically in FIG. **2**, adjacent vertical mullions **14**, horizontal head member **16** and horizontal spandrel panel support member **18** can create a frame to support building facade panel **24**, such as a spandrel cover panel to cover the spandrel area around the terminal end of a building floor slab FS.

(30) Head member **16** shown in FIGS. **2** and **4** can serve as the bottom portion of a windload connection load path of the frame above. As best shown schematically in FIGS. **2**, **2A** and **4**, head member **16** can have a top blade portion(s) **16a** configured for engaging a lower portion of the sill member **20** from the floor above. Engagement can be via rigid anchor connection to suspend or support head member **16** from sill member. An angle member **17** can be provided between or adjacent the top blade portion(s) **16a** of head member **16**. Angle member **17** can be comprised of extruded aluminum or other rigid material and have a notch to index the units above, namely downwardly projecting flanges **20a** of sill member **20**. Rigid PVC pressure spacers **80** and pressure equalization air seal gaskets **84** can be seated upon head member **18** near the terminal ends of top blade portion(s) **16a**, or along downwardly projecting flange(s) **20a** of sill members, so as to be engagingly received between top blade portion(s) **16a** of head member **16** and downward projecting flanges **20a** of sill members **20**. Pressure spacers **80** and air seal gaskets **84** and can form a seal between the horizontal head members **18** and the sill members **20** from the floor above. A vertical interior air seal **58**, such as an extruded aluminum vertical interior air seal, can be provided along the interior section of vertical mullion **14** for placement between mullion **14** and the terminal edge of floor slab FS. It will be understood that the seals formed by pressure spacers **80**, equalization air seal gaskets **84**, and vertical interior air seal **58** can have air and water-resistant capabilities.

(31) According to exemplary embodiments, air seals **82**, such as closed cell foam block air seals can be provided and sealed in place by silicone to framing members such as horizontal sill members **20** and spandrel panel support members **18**. Such air seals can provide additional insulation to the building facade system **10** to slow the transfer of heat through the system and reduce heat loss, gain and provide additional sound attenuation. As shown schematically in FIGS. **2**, **2A** and **4**, embodiments of the building facade system presented herein can include building facade insulation such as semi-rigid mineral wool **86** which is conventionally used with traditional

curtainwall systems. According to embodiments specifically presented herein, such semi-rigid mineral wool insulation **86** can be provided around portions of the frame **21** including alongside at least a portion of the vertical mullions **14** between the horizontal head members **16** and spandrel panel support members **18**. Semi-rigid mineral wool insulation **86** can also be provide within the horizontal head members **16** outside and adjacent to the body portion **34** of unified vertical shear blade anchor **32**. The semi-rigid mineral wool insulation **86** can provide additional insulation to the building facade system **10** to slow the transfer of heat through the system and reduce heat loss or gain. Weld pins **87** can be used to secure semi-rigid mineral wool insulation **86** to frame **21**.

(32) Frame seal **64** can be used to seal and secure spandrel cover panel **24** to the head member **16** and corresponding frame and a primary seal **66**, such as structural silicone and a silicone backer gasket **68** can be used to seal and secure spandrel cover panel **24** to the spandrel panel support member **18** and corresponding frame. Likewise, adjacent vertical mullions **14**, horizontal spandrel panel support member **18** and horizontal sill member **20** can support building facade panel **22**, such as an infill panel. Primary seal **66**, such as structural silicone and a silicone backer gasket **68** can be used to seal and secure infill panel **22** to the spandrel panel support member **18** and horizontal sill member **20**. It will be understood that building panels **22**, horizontal members **16**, **18** and **20**, vertical mullions **14**, unified vertical shear blade anchor **32** can be provided as a unified panel assembly **11** that can be delivered to the building site with a corresponding angled anchor member **40** for installation without the need to pre-attach anchors to the edge of the floor slab FS. Instead, panel assembly **11** can be positioned at the appropriate installation location on the building structure and angled anchor member **40** can be engaged to channel C with corresponding adjustments being made relative anchor member **40** and unified vertical shear blade anchor **32** and the anchor assembly being secured via leveling bolt anchors **60**. It will be understood that such prefabricated unified design configuration can drastically reduce installation time and costs while also enabling the placement of fire prevention measures and smoke seals.

(33) According to exemplary embodiments presented herein, infill panels **22** can be configured to extend between the spandrel cover panels **24** and enclose the building interior space between successive floor slabs FS. Infill panels can be comprised of vision glass which can be transparent, opaque, tinted, translucent, reflective and/or can be comprised of any other material selected from a group consisting of solid, perforated or patterned, steel, aluminum, glass, gfr, porcelain, sintered stone, stone and polymers. Infill panels **22** can further be insulated and/or be comprised of one or more layers and can be different dimensions or thicknesses as needed or desired. According to exemplary embodiments, spandrel cover panels **24** can be configured to extend between the infill panels **22** and cover the spandrel area around the terminal end of a building floor slab FS. Spandrel cover panels **24** can be comprised of insulating spandrel glass which can be transparent, opaque, tinted, translucent, reflective and/or can be comprised of any other material selected from a group consisting of solid, perforated or patterned, steel, aluminum, glass, gfr, porcelain, sintered stone, stone and polymers. Facade panels **22**, **24** which can be structurally glazed to frame **21** including vertical mullions **14** and horizontal members.

(34) According to exemplary embodiments shown schematically in FIG. 4, at least a portion of building facade system **10** and frame assembly **21** can be provided without horizontal sill member **20** so as to eliminate or bypass a horizontal attachment location at or around the bottom of floor slab FS and enable the use of a longer extended continuous building panel **22**. It will be recognized that such design configuration can provided a more desirable streamlined and continuous exterior aesthetic and floor to ceiling vision glass without requiring a separate spandrel cover panel adjacent the exterior edge of floor slab FS. According to exemplary embodiments shown schematically in FIG. 4, building facade system **10** can have a valance **25** made from extruded aluminum or other rigid material as desired. Valance **25** can be located inside building panel **22** so as to be positioned between building panel **22** and at least a portion of the terminal edge of floor slab FS and can provide an aesthetic benefit to hide or obscure the terminal edge of a floor slab FS. As shown

schematically in FIG. 4, valance 25 can have a cover 25a and base 25b made from extruded aluminum or other rigid material. Valance cover 25a can have a top portion that is securable to a bottom portion of horizontal head member 16, such as for example by a snap-fit connection and can extend downwardly from head member 18 at or below an elevation adjacent the bottom of floor slab FS. Valance base 25b can be secured to outside surfaces of vertical mullion 14 and can engage the lower portion of valance cover 25A. A frame seal 64, fire proofing and/or fluid applied liquid smoke seal 15 can be provided around valance base 25b as shown in FIG. 4.

(35) According to exemplary embodiments shown schematically in FIGS. 2, 2A and 4, a space can be defined between the bottom of infill panels 22 and the top of a below spandrel cover panel 24. A gasket 70 such as a rainscreen stack gasket and seal 72 such as a silicone boot seal set in a bed sealant can be set or received within such space. A weather seal and backer rod 74, extruded aluminum setting block chair and silicone glass setting block 76, and a silicone compatible perimeter thermal isolating edge adaptor 78 can be provided between the building facade panels 22, 24 and gasket 70 to seal and secure the exterior building facade against weather and have air and water-resistant capabilities.

(36) According to exemplary embodiments as best shown schematically in FIGS. 2, 2A and 4, horizontal sill member 20 can have a top portion that can extend inward from the infill panel 22 towards the building structure and cover at least a portion of an area above the flange 34 of unified vertical shear blade anchor 32. As shown schematically in FIGS. 2, 2A and 4, the top portion of horizontal sill member can have an inside edge configured for attachment of an interior trim assembly 90. Interior trim assembly 90 can have an interior trim body 92 comprising a top panel 92a and interior panel 92b. The top panel 92a of interior trim body 92 can be configured to be secured, such as for example by snap-fit connection, to the inside edge of the horizontal sill member 20 and extend inward to an opposing inside edge. As shown schematically in FIGS. 2, 2A and 4, top panel 92a can be substantially horizontal and interior panel 92b can extend substantially vertically downward from the inside edge of top panel 92a. Persons of ordinary skill in the art will recognize and appreciate that the size and shape of interior trim body 92 and/or configuration of panels 92a, 92b can be modified without limitation without departing from the novel scope of embodiments presented herein.

(37) As shown schematically in FIGS. 2, 2A and 4, interior trim assembly 90 can have a trim support member 94 and a trim index 96. Trim assembly 90 including trim body 92, trim support member 94 and trim index 96 can be made from extruded aluminum and/or other rigid materials without limitation, can be configured to be installed after the frame 21 and anchor assembly 10 are secured in place and after all positional adjustments to the anchor assembly are made. As illustrated, at least a portion of trim support member 94 can extend downward and away from the top panel of trim body 92 and substantially parallel to interior panel 92b. Trim index 96 can be provided between trim support member 94 and interior panel 92b and a gasket 98 such as a friction or compression gasket can be seated between inside edges of trim support member 94 and interior panel 92b. According to exemplary embodiments best shown schematically in FIGS. 2, 2A and 4, trim index 96 can be vertically adjustable and can be slidably engaged through gasket 98 to seal an opening or space between the top surface of the floor slab FS and the terminal end of the interior trim panel 92b. Interior finish gaskets 98 and trim index 96 can be provided to accommodate incidental building movements and concrete tolerances. According to exemplary embodiments shown schematically in the figures, trim assembly 90 including trim body 92, trim support member 94, trim index 96 and gaskets 98 can be removable to enable access to blade anchors 25, 29, angled anchor member 23 and fasteners 22 if desired.

(38) As shown schematically in FIGS. 2-4, exemplary embodiments of the facade system 10 can comprise fire and smoke-resistant seals and insulation to further prevent fire and smoke from spreading between floors. According to exemplary embodiments shown schematically in FIGS. 2-4, fire safing 26 can be provided along the interior notched section 15 of vertical mullion 14 for

placement between the interior side of mullion and the terminal edge of floor slab FS. Such fire safing **15** can better prevent the spread of smoke and fire through the space between the frame and the exterior of the building structure so that fire is less able to spread between floors. Persons of ordinary skill in the art will recognize and appreciate that the unified vertical shear blade anchor **32** in cooperation with the notched mullion **14** can enable the use of fire safing **15** as shown and described herein which cannot be provided with conventional curtainwall systems which require fire stopping measures and extra finishing after the system is installed. According to exemplary embodiments shown schematically in FIGS. 2-4, a smoke seal **15**, such as a fluid-applied liquid smoke seal can also be provided. Smoke seal **15** can be provided above at least a portion of fire safing **15** forming a seal between an interior portion of horizontal head member and the top of channel C so as to seal the space between the interior edge of vertical mullion **14** and the terminal edge of slab from the space above the floor slab FS. Together with fire safing **15**, the smoke seal **13** can prevent the spread of smoke between floors through the space between the frame and exterior of the building.

(39) Utilization of the unified vertical shear blade anchor **32** in accordance with embodiments described herein can enable the building facade system to conform to building fire code requirements calling for a traditional two (2) hour rated fire stop and smoke seal. The invention and utilization of a site indexable floor slab interface trim in accordance with the system described herein permits site adjustability of the trim to cover the gap left at the terminal top face of the slab by concrete that is not uniform without the use of an unsightly caulk joint as is required by other notched vertical curtainwalls. The unified vertical shear blade anchor in accordance with disclosed embodiments further allows for fewer installers to complete the enclosure of the building structure by eliminating the need for a separate plurality of parts to be added to the terminal end of the floor slab as generally required in traditional curtainwall systems.

(40) According to exemplary embodiments shown schematically in FIG. 8, a method **100** of forming a building facade system is provided herein. As shown in FIG. 8, the method can comprise providing **102** a unified vertical shear blade anchor having a body portion and a flange extending horizontally therefrom in a first direction. According to exemplary embodiments, the flange can have opposing top and bottom surfaces with the bottom surface having downwardly projecting serrations along at least a portion thereof. The serrations can extend in a second direction substantially perpendicular to the first direction. According to exemplary embodiments, the method can comprise forming **104** a unified pre-installed panel assembly by coupling **106** the unified vertical shear blade anchor to a vertical mullion, coupling **108** the vertical mullion to at least one horizontal support member to form a frame assembly and structurally glazing **110** a building panel to the frame assembly. As shown schematically in FIG. 8, the method **100** can comprise providing **112** an angle member having first and second flanges each having proximal ends joined together and opposing terminal ends. The first and second flanges can extend substantially perpendicular to one another with the second flange having a top surface with upwardly projecting serrations along at least a portion thereof. The upwardly projecting serrations can extend in the second direction. The method **100** can further comprise securing **114** the angle member to a building floor slab and engaging **116** the unified vertical shear blade anchor of the pre-installed panel assembly to the angle member so as to couple the pre-installed panel assembly to the floor slab. The engagement between the angle member and unified vertical shear blade anchor can be made by securing a fastener through the flange of the vertical shear blade anchor and the second flange of the anchor member and engaging at least some of the downwardly projecting serrations of the vertical shear blade anchor with at least some of the upwardly projecting serrations of the anchor member.

(41) As shown schematically in FIG. 8, the method **100** can also comprise securing **106** the unified vertical shear blade anchor by shear connection to a vertical mullion configured for supporting horizontal members. According to exemplary embodiments, the vertical mullion and horizontal members can comprise a frame for supporting building facade panel to form the building facade.

Securing of the vertical shear blade anchor by shear connection to the vertical mullion can comprise coupling the body portion of the vertical shear blade anchor to an outside lateral surface of the vertical mullion. According to exemplary embodiments, such coupling can comprise the use of a fastener extending in the second direction through a portion of the body portion and vertical mullion. According to exemplary embodiments shown schematically in FIG. 8, the method **100** can also comprise securing **108** the horizontal members to the vertical mullion to form the frame and securing **110** the building facade panel to the frame by way of structural glazing for example. The assembly when fastened together can provide a compressioned anchor for a building facade system.

(42) Methods according to exemplary embodiments shown schematically in FIG. 8 can further comprise applying **117** fire safing along an interior notched section **15** of vertical mullion **14** between the interior side of mullion and the terminal edge of floor slab FS and applying **119** a smoke seal such as a fluid-applied liquid smoke seal, above at least a portion of the firesafing between an interior side of the horizontal head member and the top surface of channel C. The smoke seal can form a seal between the horizontal head member and building floor slab.

(43) According to exemplary embodiments shown schematically in FIG. 8, exemplary methods can further comprise providing **118** a floor closure sub-assembly having a vertically adjustable interior trim angle held in place by compression of adjacent gaskets wherein the adjustable interior trim angle is slidably adjustable in a substantially vertical direction to interface an interior finish of the building floor slab. Methods provided herein can comprise securing **120** the floor closure sub-assembly to at least one of the horizontal members of the unified pre panel assembly to provide an interior trim assembly for the building facade system. According to exemplary embodiments, the method can further include sealing **122** an opening or space between the top surface of the floor slab FS or interior floor surface and the interior trim assembly by slidably adjusting an interior trim index in a substantially vertical direction towards the floor slab.

(44) From the foregoing, it will be observed that numerous variations and modifications may be affected without departing from the spirit and scope of the invention. It is to be understood that no limitation with respect to the specific apparatus illustrated herein is intended or should be inferred. It is, of course, intended to cover by the appended claims all such modifications as fall within the scope of the claims. From the foregoing, it will be seen that this invention is one well adapted to attain all the ends and objects hereinabove set forth together with other advantages which are inherent to the structure. It will be understood that certain features and sub combinations are of utility and may be employed without reference to other features and sub combinations. Since many possible embodiments of the invention may be made without departing from the scope thereof, it is also to be understood that all matters herein set forth or shown in the accompanying drawings are to be interpreted as illustrative and not limiting.

(45) Further, logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. Other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from the described embodiments.

Claims

1. A building facade system comprising: a frame assembly for supporting a building facade panel; an anchor having a body portion and a flange extending horizontally in a first horizontal direction from the body portion, the flange having serrations along at least a portion thereof; a vertical mullion secured by a shear connection to the anchor, the vertical mullion being held in shear and being configured for supporting horizontal members to form the frame assembly for supporting the building facade panel, the shear connection between the vertical mullion and the anchor being formed by a fastener coupling the anchor to an outside lateral surface of the vertical mullion, the fastener extending in a second horizontal direction substantially perpendicular to the first horizontal

- direction through at least a portion of the body portion and vertical mullion; and an angle member having first and second flanges each having proximal ends joined together and opposing terminal ends, the first and second flanges extending substantially perpendicular to one another, the second flange being configured for engagement with the flange of the anchor by way of a vertically-oriented member coupling the flange of the anchor and the angle member, the angle member being securable to a building floor slab and supporting the anchor thereon.
2. The building facade system of claim 1 wherein the anchor comprises a plurality of spaced apart anchors, each of the plurality of spaced apart anchors being separately secured by shear connection to an outside lateral surface of the vertical mullion suspending the vertical mullion in shear.
 3. The building facade system of claim 1 wherein the vertical mullion comprises a plurality of vertical mullions each formed from complementary mullion halves.
 4. The building facade system of claim 1 further comprising the building facade panel, wherein the building facade panel comprises at least one of a slab edge cover and an infill panel.
 5. The building facade system of claim 4 wherein the building facade panel comprises a plurality of building facade panels.
 6. The building facade system of claim 1 further comprising the horizontal members, the horizontal members comprising at least one of a split and hollow horizontal member fastened in shear to said vertical mullion to create the frame assembly.
 7. The building facade system of claim 4 wherein the infill panel is attached to the frame assembly to create at least one of a watertight and non-watertight barrier.
 8. The building facade system of claim 1 further comprising at least one of site-installed fire safing material and a site-installed siliconized smoke seal, the fire safing material being applied along at least a portion of an interior section of the vertical mullion and a terminal edge of the building floor slab, the smoke seal being applied above at least a portion of the fire safing to seal a space between the interior section of the vertical mullion and the terminal edge of the building floor slab.
 9. The building facade system of claim 1 further comprising a floor closure sub-assembly mechanically affixed to a top portion of at least one of the horizontal members, the floor closure sub-assembly comprising a vertically adjustable interior trim angle wherein the adjustable interior trim angle is slidably adjustable in a substantially vertical direction to interface an interior finish of the building floor slab.
 10. The building facade system of claim 1 wherein the flange of the anchor comprises a slotted opening extending therethrough, the slotted opening having a length extending horizontally in the first direction to permit horizontal adjustment of the building facade system.
 11. The building facade system of claim 1 wherein the anchor, vertical mullion, horizontal members and building facade panel are coupled together before being installed on a building structure and anchored to the building structure together as a unified panel assembly.
 12. The building facade system of claim 4 wherein the infill panel is structurally glazed to the frame assembly and comprised of a material selected from a group consisting of solid, perforated or patterned, steel, aluminum, glass, glass fiber reinforced concrete, porcelain, sintered stone, stone and polymers.
 13. The building facade system of claim 8 wherein the site-installed fire safing material comprises mineral wool.
 14. The building facade system of claim 4 wherein the infill panel is mechanically secured to the frame assembly and comprised of a material selected from a group consisting of solid, perforated or patterned, steel, aluminum, glass, glass fiber reinforced concrete, porcelain, sintered stone, stone and polymers.
 15. The building facade system of claim 1 wherein the vertically-oriented member coupling the flange of the anchor and the angle member is adjustable to permit vertical adjustment of the building facade system.
 16. The building facade system of claim 12 wherein the infill panel creates at least one of a

watertight and non-watertight barrier with the frame assembly.

17. The building facade system of claim 14 wherein the infill panel creates at least one of a watertight and non-watertight barrier with the frame assembly.

18. A building facade system comprising: a frame assembly for supporting a building facade panel; an anchor having a body portion and a flange extending horizontally in a first horizontal direction from the body portion, the flange having serrations along at least a portion thereof; a vertical mullion secured by a shear connection to the anchor, the vertical mullion being held in shear and being configured for supporting horizontal members to form the frame assembly for supporting the building facade panel, the shear connection between the vertical mullion and the anchor being formed by a fastener coupling the anchor to an outside lateral surface of the vertical mullion, the fastener extending in a second horizontal direction substantially perpendicular to the first horizontal direction through at least a portion of the body portion and vertical mullion; an angle member having first and second flanges each having proximal ends joined together and opposing terminal ends, the first and second flanges extending substantially perpendicular to one another, the second flange being configured for engagement with the flange of the anchor by way of a vertically-oriented member coupling the flange of the anchor and the angle member, the angle member being securable to a building floor slab and supporting the anchor thereon; and wherein the building facade panel comprises at least one of a slab edge cover and an infill panel.

19. The building facade system of claim 18 wherein the anchor comprises a plurality of spaced apart anchors, each of the plurality of spaced apart anchors being separately secured by shear connection to an outside lateral surface of the vertical mullion suspending the vertical mullion in shear.

20. The building facade system of claim 18 wherein the vertical mullion comprises a plurality of vertical mullions each formed from complementary mullion halves.

21. The building facade system of claim 18 wherein the building facade panel comprises a plurality of building facade panels.

22. The building facade system of claim 18 further comprising the horizontal members, the horizontal members comprising at least one of a split and hollow horizontal member fastened in shear to said vertical mullion to create the frame assembly.

23. The building facade system of claim 18 wherein the infill panel is attached to the frame assembly to create at least one of a watertight and non-watertight barrier.

24. The building facade system of claim 18 further comprising at least one of site-installed fire safing material and a site-installed siliconized smoke seal, the fire safing material being applied along at least a portion of an interior section of the vertical mullion and a terminal edge of the building floor slab, the smoke seal being applied above at least a portion of the fire safing to seal a space between the interior section of the vertical mullion and the terminal edge of the building floor slab.

25. The building facade system of claim 18 further comprising a floor closure sub-assembly mechanically affixed to a top portion of at least one of the horizontal members, the floor closure sub-assembly comprising a vertically adjustable interior trim angle wherein the adjustable interior trim angle is slidably adjustable in a substantially vertical direction to interface an interior finish of the building floor slab.

26. The building facade system of claim 18 wherein the flange of the anchor comprises a slotted opening extending therethrough, the slotted opening having a length extending horizontally in the first horizontal direction to permit horizontal adjustment of the building facade system.

27. The building facade system of claim 18 wherein the anchor, vertical mullion, horizontal members and building facade panel are coupled together before being installed on a building structure and anchored to the building structure together as a unified panel assembly.

28. The building facade system of claim 18 wherein the infill panel is structurally glazed to the frame assembly and comprised of a material selected from a group consisting of solid, perforated or

patterned, steel, aluminum, glass, glass fiber reinforced concrete, porcelain, sintered stone, stone and polymers.

29. The building facade system of claim 24 wherein the site-installed fire safing material comprises mineral wool.

30. The building facade system of claim 18 wherein the infill panel is mechanically secured to the frame assembly and comprised of a material selected from a group consisting of solid, perforated or patterned, steel, aluminum, glass, glass fiber reinforced concrete, porcelain, sintered stone, stone and polymers.

31. The building facade system of claim 18 wherein the vertically-oriented member coupling the flange of the anchor and the angle member is adjustable to permit vertical adjustment of the building facade system.

32. The building facade system of claim 28 wherein the infill panel creates at least one of a watertight and non-watertight barrier with the frame assembly.

33. The building facade system of claim 30 wherein the infill panel creates at least one of a watertight and non-watertight barrier with the frame assembly.

34. A building facade system comprising: an anchor having a body portion and flange extending horizontally in a first horizontal direction from the body portion, the flange having serrations along at least a portion thereof, the anchor being securable to a vertical support member of a building facade panel frame assembly by way of a shear connection whereby the vertical support member is held in shear and is configured for supporting horizontal members to form the frame assembly, the shear connection between the vertical support member and the anchor being formed by a fastener coupling the anchor to an outside lateral surface of the vertical support member, the fastener configured to extend in a second horizontal direction substantially perpendicular to the first horizontal direction through at least a portion of the body portion and vertical support member; and an angle member having first and second flanges each having proximal ends joined together and opposing terminal ends, the second flange being configured for engagement with the flange of the anchor by way of a vertically-oriented member coupling the flange of the anchor and the angle member, the angle member being securable to a building floor slab and supporting the anchor thereon.

35. The building facade system of claim 34 wherein the anchor comprises a plurality of spaced apart anchors, each of the plurality of spaced apart anchors being separately secured by shear connection to an outside lateral surface of the vertical support member suspending the vertical support member in shear.

36. The building facade system of claim 34 wherein the vertical support member comprises a plurality of vertical support members each formed from complementary halves.

37. The building facade system of claim 34 further comprising the building facade panel, wherein the building facade panel comprises at least one of a slab edge cover and an infill panel.

38. The building facade system of claim 37 wherein the building facade panel comprises a plurality of building facade panels.

39. The building facade system of claim 34 further comprising the horizontal members, the horizontal members comprising at least one of a split and hollow horizontal member fastened in shear to said vertical support member to create the frame assembly.

40. The building facade system of claim 37 wherein the infill panel is attached to the frame assembly to create at least one of a watertight and non-watertight barrier.

41. The building facade system of claim 34 further comprising at least one of site-installed fire safing material and a site-installed siliconized smoke seal, the fire safing material being applied along at least a portion of an interior section of the vertical support member and a terminal edge of the building floor slab, the smoke seal being applied above at least a portion of the fire safing to seal a space between the interior section of the vertical support member and the terminal edge of the building floor slab.

42. The building facade system of claim 34 further comprising a floor closure sub-assembly mechanically affixed to a top portion of at least one of the horizontal members, the floor closure sub-assembly comprising a vertically adjustable interior trim angle wherein the adjustable interior trim angle is slidably adjustable in a substantially vertical direction to interface an interior finish of the building floor slab.
43. The building facade system of claim 34 wherein the flange of the anchor comprises a slotted opening extending therethrough, the slotted opening having a length extending horizontally in the first horizontal direction to permit horizontal adjustment of the building facade system.
44. The building facade system of claim 34 wherein the anchor, vertical support members, horizontal members and building facade panel are configured to be coupled together before being installed on a building structure and configured to be anchored to the building structure together as a unified panel assembly.
45. The building facade system of claim 37 wherein the infill panel is structurally glazed to the frame assembly and comprised of a material selected from a group consisting of solid, perforated or patterned, steel, aluminum, glass, glass fiber reinforced concrete, porcelain, sintered stone, stone and polymers.
46. The building facade system of claim 41 wherein the site-installed fire safing material comprises mineral wool.
47. The building facade system of claim 37 wherein the infill panel is mechanically secured to the frame assembly and comprised of a material selected from a group consisting of solid, perforated or patterned, steel, aluminum, glass, glass fiber reinforced concrete, porcelain, sintered stone, stone and polymers.
48. The building facade system of claim 34 wherein a vertically-oriented member coupling the flange of the anchor and the angle member is adjustable to permit vertical adjustment of the building facade system.
49. The building facade system of claim 45 wherein the infill panel creates at least one of a watertight and non-watertight barrier with the frame assembly.
50. The building facade system of claim 48 wherein the infill panel creates at least one of a watertight and non-watertight barrier with the frame assembly.
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